

JENNISHA MARTIN

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SUMMARY

Data Engineer with 3+ years of professional experience designing large-scale ETL pipelines, dimensional data models, and distributed data architectures across healthcare and enterprise domains. Proficient in Python, SQL, PySpark, Hadoop/HDFS, Node.js, and AWS, with applied NLP work using SpaCy, Scikit-Learn, and PyTorch. Skilled in integrating ML models into data workflows, building reliable batch and real-time pipelines, improving data quality, and providing teams with clean, accessible, and business ready data.

SKILLS

Languages: Python, SQL, Java, C, C++, JavaScript, Bash

Databases: MySQL, PostgreSQL, Snowflake, Redshift, SQL Server, Azure SQL DB, MongoDB

ETL & Big Data: Airflow, PySpark, Hadoop, HDFS, Databricks, dbt, AWS Glue, Snowpipe, Data Modeling

Cloud & DevOps: AWS (S3, EC2, Lambda), Terraform, Docker, Kubernetes, Jenkins, CI/CD, Git

AI/ML: Scikit-Learn, SpaCy, PyTorch, LangChain, RAG, Langfuse, LLM Evaluation, LiteLLM, Pinecone

BI & Analytics: Power BI, Tableau, Looker Studio, DAX, SSAS, SSRS, Power Query, KPIs, A/B Testing

EXPERIENCE

IpsierLab | *AI Engineer Intern* | Remote, Texas, USA

Sep 2025 – Dec 2025

- Built an AI voice assistant for mortgage prequalification using **LiveKit**, **Deepgram**, and **Twilio SIP**, enabling real-time borrower interactions and agent-to-human transfer.
- Designed **LangGraph** workflows with **NLP** intent classification (**SpaCy**, **Scikit-Learn**, **PyTorch**) for borrower intake and lead routing, using **PostgreSQL** checkpointing for multi-turn state.
- Productionized ML models into **RAG** pipelines using **LangChain**, **FastAPI**, and **Node.js**, integrating **Langfuse**, **Prometheus**, and **Grafana** for observability and latency monitoring.

Infosys Limited | *Data Engineer* | Chennai, India

Aug 2021 – Dec 2023

- Built large-scale **ETL** pipelines with **AWS Glue**, **PySpark**, **HDFS**, and **SQL** to ingest **10M+** ERP records from **XML**, **JSON**, and relational sources into **Snowflake**, cutting processing time by **20%**.
- Modeled dimensional schemas and **data marts** in **Redshift** powering **25+** real-time and batch workflows, enforcing **data quality, lineage, and accessibility standards** across **10+** sources.
- Optimized **50+** SQL queries and stored procedures in **PostgreSQL** and **Amazon Redshift**, reducing average query execution time by **20%** for critical business intelligence and executive reporting.
- Orchestrated **15+** **Airflow** pipelines with **Docker** and **Jenkins CI/CD**, and partnered with **Data Science** to integrate ML feature outputs into BI workflows for downstream stakeholder reporting.
- Developed **25+** **Tableau** dashboards for Finance, Marketing, and Sales leadership tracking revenue, pipeline, churn, and campaign ROI, helping teams monitor business performance more effectively.

Infosys Limited | *Data Engineer Intern* | Mysore, India

Feb 2021 – Jun 2021

- Engineered a marketing analytics pipeline using **Python** and **PostgreSQL**, replacing manual spreadsheet workflows with a centralized database serving **5+** marketing teams, reducing reporting effort by **30%**.
- Optimized **PostgreSQL** queries to improve performance by **15%** and established a **MongoDB**-based meta-data store for audit logging and data lineage tracking across **10+** data sources.
- Created **Power BI** reports tracking campaign ROI, conversion rates, and engagement trends, enabling marketing stakeholders to identify underperforming channels and improve campaign efficiency by **20%**.

PROJECTS

AutoMend | *Airflow, Ray, Polars, Docker, DVC, BERT, Llama-3, Python*

Jan 2026 – Apr 2026

- Built an autonomous **MLOps** remediation platform (**3rd place, Google Boston**) using **BERT** and **Llama-3** models to detect and resolve ML incidents across **6** datasets, cutting response time by **40%**.
- Orchestrated pipelines via **Airflow** with **Ray** distributed processing, **Polars** validation, **DVC** versioning, **Fairlearn** bias detection, and Slack alerting, containerized in **Docker**.

SupplyFlow | *AWS S3, Lambda, Glue, Redshift, PySpark, Power BI*

Jan 2025 – Mar 2025

- Architected a serverless AWS **ETL** pipeline using **S3**, **Lambda**, **Glue**, and **Redshift** to transform supply-chain and geospatial data into a **9-table** snowflake-schema warehouse.
- Developed **Power BI** dashboards monitoring profitability, carrier performance, and delivery risk with drill-through for strategic decisions.

- HealthSync

| Python, SQL, dbt, Snowflake, AWS S3, Looker Studio

Sep 2024 – Nov 2024
- Constructed an end-to-end **healthcare data pipeline** ingesting **CMS Medicare claims** into a unified **Snowflake** star schema using **medallion architecture**, **dbt** models, and **SQL** transformations.
 - Identified opioid overprescription and duplicate-claim patterns using **ICD and HCPCS** code mappings, and delivered **Looker Studio** dashboards for billing outliers, cost benchmarking, and provider analytics.

- TweetPulse

| Python, Azure Blob Storage, EventGrid, Snowpipe, Snowflake

Jan 2024 – Mar 2024
- Deployed a Python-based **real-time ingestion** pipeline with **Azure Blob Storage** to stream Twitter feeds into **Snowflake** via **Snowpipe** with cross-account IAM integration.
 - Designed a **Snowflake** dashboard with **30-minute** auto-refresh for tweet engagement metrics, trending topics, and sentiment patterns.

ACHIEVEMENTS

- 3rd Place – Google, Cambridge, Boston

Apr 2026

Secured third place for AutoMend, a self-healing MLOps platform leveraging BERT and Llama-3 models for autonomously remediating ML incidents.
- Infosys Rising Star Award, Infosys Limited

Apr 2023

Awarded for exemplary performance, initiative in developing features, and improving reporting performance.
- Infosys Insta Award, Infosys Limited

Apr 2022

Recognized for outstanding contributions and demonstrating strong growth potential as an early Infosys employee.

EDUCATION

Northeastern University, Boston, MA
Masters in Computer Science – Khoury College of Computer Sciences

Anna University, Chennai, India
Bachelors in Computer Science and Engineering